



CVR: An Automated CV Recommender System Using Machine Learning Techniques

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Abstract. In the recruitment process, hand-picking the right candidate out of a pool of resumes in the allotted time can be very challenging. Furthermore, as there is no standard pattern for producing CVs, there is always the risk of overlooking crucial information. This research focuses on extracting important information from CVs and job descriptions of various formats, using several machine learning techniques. The extracted data from CVs and job descriptions is then compared to find the best-matched CVs using a variety of metrics such as cosine similarity, soft cosine similarity, Jaccard similarity, dice similarity coefficient, overlap coefficient, and conditional probability. Conditional probability outperforms other metrics and hence is chosen as the metric to assess similarity. Following this, the content-based recommendation technique is used to recommend candidates based on skillset. Currently, our work is limited to computer science, but we plan to extend our work into other sectors in the future.

Keywords: CV · Job description · Recommendation · Similarity · Skill

1 Introduction

The introduction of online job sites (such as LinkedIn, Indeed, bdjobs, etc.) in recent years has significantly enhanced the process of attracting job applicants [1]. These job boards have made it easier for candidates to apply for jobs, but they have also presented new challenges for recruiters. According to a recent survey, 118 people apply for a single job on average, with only 24 being summoned for a normal interview [2]. It can be stressful for recruiters to shortlist

potential individuals for a job position within a limited time when there are so many applicants. Moreover, CV templates can vary from one job applicant to another. This, sometimes, results human resources (HR) to unintentionally overlook important elements in a CV [3]. To address this, many organizations specify the structure for producing a CV [4]. This partial solution often makes it difficult for candidates to express themselves within a fixed constrained format.

Considering these challenges, this work aims to assist HR by developing an automated system for screening out skilled candidates fit for the job profile, regardless of their CV format. To achieve this goal, CVs and job descriptions (JDs) were initially obtained from several sources. Following that, a parser was built with the help of machine learning techniques to extract relevant information from the collected CVs and job descriptions. Following this, six similarity checking algorithms have been applied to match the skills presented in the CVs to the job requirements. Finally, the similarity scores obtained from the conditional probability algorithm are employed in a content-based recommendation system to present the top N CVs to recruiters. In the end, a web-based application is developed to deploy the system.

The rest of the sections are organized as follows: In Sect. 2, detailed descriptions of similar works are presented. The methodology of constructing the automated CV recommendation system is provided in Sect. 3. Section 4 describes the findings and results of this work, Finally, the concluding remark is given in Sect. 5.

2 Related Work

Recently, research in the fields of artificial intelligence has seen a variety of applications in HR [5–7]. The rise of e-recruitment has resulted in a considerable shift in the hiring process. Several research offered several methodologies with the goal of automating candidate screening [8]. Bhor et al. [9] presented a comprehensive survey that includes a job site where employees and applicants can upload their resumes for any job opening. The NLP technique graded employee resumes based on the required skill set of the organization and the skills listed in the person's resume. Roy et al. [10] discussed an automated supervised machine learning-based system that recommends appropriate applicant resumes to HR based on the job description. Content-based with cosine similarity and KNN were applied as a recommendation model. The model takes CVs in CSV format, and uses the genism library.

Arunasish and colleagues' "SCREENER" model [11] scans resumes for essential features of candidate profiles such as skills, experience with each skill, education, and past work experience and parses them using Apache Tika.

Daryani et al. [12] presented an automated resume screening system that employs NLP to extract relevant information from unstructured resumes, such as skills, education, and experience, and then tokenizes each application to construct a summary form. After deleting unnecessary data, the applicants were rated using a content-based recommendation that matches the extracted resume

attributes to the job need using the Vector Space Model and cosine similarity. Using a cosine similarity method based on verbal job description, Agarwal and Senthilkumar [13] created an autonomous machine learning model that assisted in recommending eligible candidates' resumes to recruiters. Al-Otaibi and Ykhlef [14] used three similarity measures to rank candidates for a job position. They described a survey-based job recommender system in which content-based filtering works effectively with implicit feedback when explicit rating is impossible. Resumes were classified using the KNN algorithm, and cosine similarity was used to determine how close the candidate's resume was to the job description. Rajath et al. [15] employed KNN Algorithm to categorize resumes according to their respective categories, while cosine similarity was used to determine how similar is the candidate's CV to the job description.

Kiran et al. [16] used data mining techniques to discover patterns that can aid in profession and skill suggestions. Apriori algorithm was to create association rules and recommendations. Gupta and Garg [17] used content-based profile matching to recommend jobs to aspiring candidates. Kumar [18] used Naive Bayes to assess resumes. The accepted candidates were contacted with an offer message whereas the rejected candidates receive appropriate feedbacks. Wang et al. [19] introduced a knowledge graph and BERT-based keyword-based search engine for recruitment and staffing. Gugnani and Misra [20] proposed a system for job suggestion in which skills were extracted using NLP techniques and Doc2VecModel and similarity approaches were utilized to discover and infer implicit abilities for each job description that were not explicitly listed in the original job description. A generalizable ensemble method for extracting abilities from unstructured material in resumes and JDs was also proposed in the paper. Duan et al. [21] proposed a personalized resume recommendation algorithm. They extracted the features and used K-means++ to cluster position, build text vectors, and combine TF-IDF and part-of-speech weights to recommend resumes using cosine similarity. Moreover, they proved that the proposed algorithm's recommendation result outperforms the word frequency vector. Appadoo and colleagues [22] presented the development of JobFit, a job recommendation system that uses various ML models, a collaborative filtering recommender system, and past data to predict the best fit candidate for a job. The proposed job recommendation system takes the job requirement and applicant profile as inputs and outputs a JobFit score indicating how to fit each applicant for the particular job. Kurdija et al. [23] described a CV recommender that used a content-based filtering system and divided it into two properties - the ability to classify candidates into roles based on the automatic processing of their CV documents and the ability to recommend missing skills based on various similarity-based strategies to make the candidate's CV more accurate and complete.

3 Methodology

Different job sectors have different requirements and as a consequence CVs for different job sectors often have different styles and formats. In this research, we

restrict our work to the field of *computer science*. Since there exists a lot of variation in CVs it is impossible to build any rule based solution to find the right CVs against JDs. Therefore, to achieve the goal of proposing appropriate CVs, a machine learning-based approach has been taken. Figure 1 shows the workflow adopted in this study which can be broadly divided into six key segments including data acquisition, data preprocessing, data extraction, similarity checking, recommending top N CVs and deploying the model as a web app.

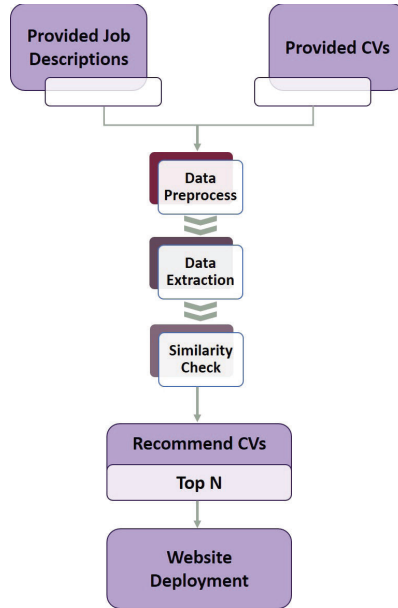


Fig. 1. Methodology

3.1 Dataset

The objective of this work is to create a model that will be able to recommend the right CVs against a given job description. In order to realise it, multiple CVs with corresponding job descriptions were collected (dataset). Individuals were contacted via social media as well via word of mouth and were given a brief idea about this research. Individuals, who are interested to participate in this research, were requested to supply their current CVs. Job descriptions were accepted in PDF as well as in text formats, while CVs were only accepted in PDF format. A total of 293 CVs were initially collected. Some CVs were in image formats and for these CVs no information could be gleaned. These CVs were removed, leaving a total of 181 CVs to be utilized as references in the parser's construction. These 181 CVs differ in terms of layouts, writing style,

and length. Personal information, educational background, skills, professional experience, and so on are all reflected uniquely in each CV. Furthermore, the length of the CVs received spans from 1 page to 4 pages. However, the bulk of CVs are two pages long, accounting for around 45% of all CVs used.

3.2 Data Preprocessing

After taking the inputs, PDFs were converted into text for data preprocessing. To start with the preprocessing steps, regular expressions [24] and some built-in python functions were used to remove brackets, HTML tags, and unnecessary whitespaces from job description. Similar approaches were applied on CVs to get clean data (e.g., phone no. and skills). Following this, word tokenization and sentence tokenization [12] were used on both. Figure 2 shows the raw job description along with the tokenized one. Moreover, the stopwords removal [10] had been applied to these tokenized words to remove unnecessary words that occur frequently but do not add any values. In addition to these, all texts were converted to lower case. For further preprocessing, stemming and lemmatization were applied to get the base form of each tokenized words. Stemming is the process of reducing a word to its word stem that affixes to suffixes and prefixes or the roots of words known as a lemma [10]. Lemmatization is closely related to stemming. It is the process of grouping together the different inflected forms of a word into the root form, which has the same meaning [10]. WordNetLemmatize, a class of NLTK, was used to lemmatize the job description. Furthermore, the Porter Stemmer algorithm [25] was employed to remove the commoner morphological and inflectional endings from words in English. The primary usage of this is a part of the normalization process usually done when setting up information retrieval systems [25]. In this work, lemmatization gives better result than stemming technique.

```
Raw Job Description
Job Requirements: Strong understanding of fundamentals of programming, data structures
and algorithms Strong problem-solving skills Hands-on experience with at least 1 prog
ramming language and its eco-system e.g. Javascript Node/Browser, C# ASP.NET/WinForms,
Java Android/Spring, etc. Hands-on experience with any SQL or NoSQL database Educationa
l Qualifications: Bachelor in Computer Science or any other department but having good
programming skills. Employment Status: Full-time

Tokenized Job Description
['Job', 'Requirements', 'Strong', 'understanding', 'of', 'fundamentals', 'of', 'program
ming', 'data', 'structures', 'and', 'algorithms', 'Strong', 'problem', 'solving', 'skil
ls', 'Hands', 'on', 'experience', 'with', 'at', 'least', 'i', 'programming', 'languag
e', 'and', 'its', 'eco', 'system', 'e', 'g', 'Javascript', 'Node', 'Browser', 'C', 'AS
P', 'NET', 'WinForms', 'Java', 'Android', 'Spring', 'etc', 'Hands', 'on', 'experience',
'with', 'any', 'SQL', 'or', 'NoSQL', 'databaseEducational', 'Qualifications', 'Bachelo
r', 'in', 'Computer', 'Science', 'or', 'any', 'other', 'department', 'but', 'having',
'good', 'programming', 'skills', 'Employment', 'Status', 'Full', 'time']
```

Fig. 2. Tokenized job description

3.3 Data Extraction

After applying the preprocessing steps, data extraction is done on CVs and job description. To do so, a parser is built to extract information using regular expression, Natural Language Processing (NLP) and the Natural Language Toolkit (NLTK).

Firstly, the name of the candidate is extracted from the name of the PDF as a specific PDF naming format is used. Then, regular expression and some python built-in functions were used to extract contact numbers and emails of candidates.

In addition, two datasets for extracting skills and degrees were produced. One has Computer Science-related talents, while the other provides degree names (e.g., BSc, MSc etc.). Using the skillset dataset on the tokenized words, skills from CVs and job description are extracted. However, the n-gram (bigram and trigram) of NLP which is the consecutive sequence of words in a text is used to grab the skills containing more than one word [26]. Moreover, obtained degrees and highest degree of candidates are extracted from the CVs that matched with the created degree names database. Then word formatting has been done on the extracted skills and degree names so that differently written skills and degrees in different places become identical. It makes the similarity checking process easier.

For further extraction, POS Tagging is applied on the tokenized words of CVs. It is the process of NLTK for marking up words in text format for a specific part of speech depending on their meaning and context [12]. From these tagged words, CGPA of the candidates are extracted from the CVs. Lastly, another information, experience details of candidates have also been extracted in the developed system. To do so, a NLP based python package, resume-parser has been used [27].

Thus, the parser used for this system could extract the following fields from a CV shown in Fig. 3. After parsing all the CVs, the extracted information is stored in a CSV file for further usage in this recommendation system.

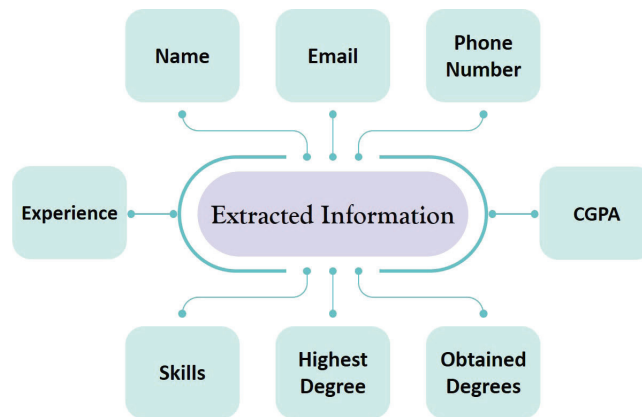


Fig. 3. Extracted fields after parsing

3.4 Similarity Check

In this phase, the similarity between the job description and the CVs has been checked. Several approaches have been taken for this including that of Cosine similarity, Soft Cosine Similarity, Jaccard Similarity, Dice Similarity Coefficient, Overlap Coefficient and Conditional Probability.

Cosine Similarity: The first similarity checker is Cosine Similarity where a word's vector representation is created, and the texts are represented in n-dimensional vector space. The skills extracted from CVs are converted to vectors using TfidfVectorizer. It converts a set of raw documents into a TF-IDF feature matrix. The abbreviation TF-IDF stands for Term Frequency and Inverse Document Frequency [12]. It is a numerical metric used to estimate the importance of a term with respect to a document in a collection or corpus [12]. Next, the CountVectorizer() is used to check the frequency of each word that appears in the job description and the CV which is shown in the Fig. 4. After that, we checked the similarities between the extracted skills from CVs and the given job description. The output of the vectorization comes as a sparse matrix [12]. The formula of Cosine Similarity is given below:

$$CS(JD, CV) = \cos(\theta) = \frac{JD \cdot CV}{||JD|| ||CV||} . \quad (1)$$

	android	angular	asp	bootstrap	css	html	javascript	net	python	sql	vue
CV	0.364693	0.000000	0.259482	0.364693	0.364693	0.364693	0.259482	0.259482	0.364693	0.364693	0.000000
JD	0.000000	0.533098	0.379303	0.000000	0.000000	0.000000	0.379303	0.379303	0.000000	0.000000	0.533098

Fig. 4. Word frequency using TF-IDF

Soft Cosine Similarity: The next one is Soft Cosine Similarity. Unlike Cosine Similarity, it considers the semantic meaning of words. A dictionary containing a word map to unique id, the corpus (word counts) for each sentence, and the similarity matrix is prepared for computing soft cosine similarity [28]. The metric used for Soft Cosine Similarity is in the following:

$$SCS(JD, CV) = \frac{\sum_{i,j}^N s_{ij} JD_i CV_j}{\sqrt{\sum_{i,j}^N s_{ij} JD_i JD_j} \sqrt{\sum_{i,j}^N s_{ij} CV_i CV_j}} . \quad (2)$$

Jaccard Similarity: Another similarity metric, Jaccard Similarity, is calculated by dividing the intersection of two texts by their union [29]. The intersection and union are found by using a set of words of texts. The formula used is given below:

$$JS(JD, CV) = \frac{|JD \cap CV|}{|JD \cup CV|} . \quad (3)$$

Dice Similarity Coefficient: The next one is Dice Similarity Coefficient. Dice Similarity Coefficient or the Sorensen-Dice index have almost the same impact as Jaccard similarity [30]. This metric is equal to twice the number of items identical by both sets divided by the total number of items in both sets as shown below:

$$DSC(JD, CV) = \frac{2|JD \cap CV|}{|JD| + |CV|} . \quad (4)$$

Overlap Coefficient: Overlap Coefficient is another technique that has been used to find similarities. The overlapping co-efficient is defined as the intersection of the skill sets of the CV and Job Description, divided by the minimum of skill set [31]. The formula of Overlap Coefficient is given below:

$$OS(JD, CV) = \frac{|JD \cap CV|}{\min(|JD|, |CV|)} . \quad (5)$$

Conditional probability: The last used metric is Conditional probability. It calculates the similarity score dividing the intersection of two texts by the text for which we are looking for the similarity as shown below [32]:

$$P(CV|JD) = \frac{P(CV \cap JD)}{P(JD)} . \quad (6)$$

The similarity score for all checkers ranges from 0 to 1. The more the score is nearer to 1, the more similar the texts are [33].

3.5 Recommending Top N CVs

After using different methods for checking similarities, the last phase of the workflow of the proposed system is to recommend top N CVs. Two techniques-K Nearest Neighbor and Content-based recommendation are applied for this.

The first one, K Nearest Neighbor (KNN) is given the best matched CV with the job description to obtain the recommended CVs. In KNN, the quantity of recommended CVs must be specified. Another applied recommendation technique, Content-based recommendation, which makes recommendations based on similarities in features, was used. This approach is employed in recommender systems, in which algorithms are developed to recommend documents to users based on gathered information [34]. In the system that we have developed, it takes the most similar CV after checking the similarity with the job description. This recommendation technique recommends top N CVs based on the similarity scores between the top CV and other CVs.

3.6 Website Deployment

A web platform (in the local machine) is developed for future widespread use of this CV recommendation system. One of the key components of this website is the “Find Candidate” component. To use this and get CV recommendations,

	Name	Email	Phone_no	Skills	Obtained Degrees
38	KARNISH KHANAM	karnishalu@gmail.com	01990198144	angular,arduino,asp.net,asp.net core,bitbucket...	[bachelor]
75	MD. FORMAN ULLAH SAJIB	formanullahsajib@gmail.com	8801521484327	android,angular,angular cli,artificial intelli...	[b.sc', 'master', 'ms']
78	MD. IMRAN HOSSAIN	imran125651@gmail.com	01518475651	asp.net,asp.net core,bootstrap,c,html,jquery,m...	['ssc', 'hsc']
160	SHUVO DAS	Shuvod308@gmail.com	8801680998152	android,angular,arduino,asp.net,asp.net core,a...	['b.sc', 'bachelor', 'ssc', 'ms', 'hsc']
42	KOUSHIK KUMAR DAS	koushikdas19966@gmail.com	01796205929	arduino,artificial intelligence,bootstrap,c,co...	[b.sc]

Fig. 5. Content based recommendation

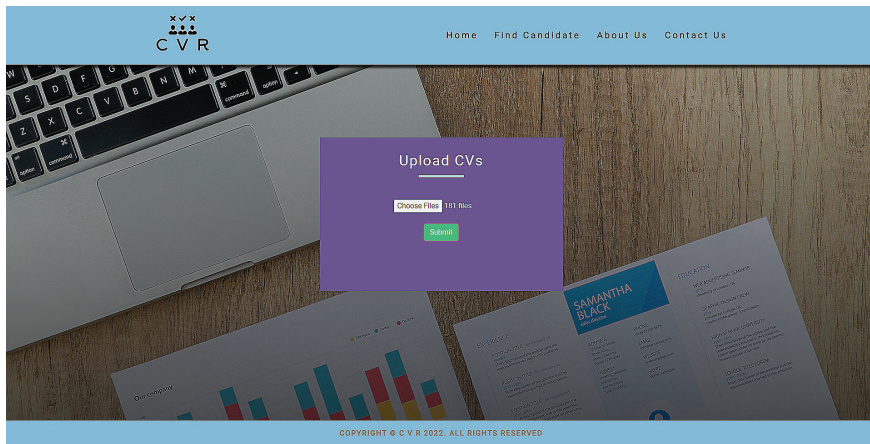


Fig. 6. Interface of uploading CVs

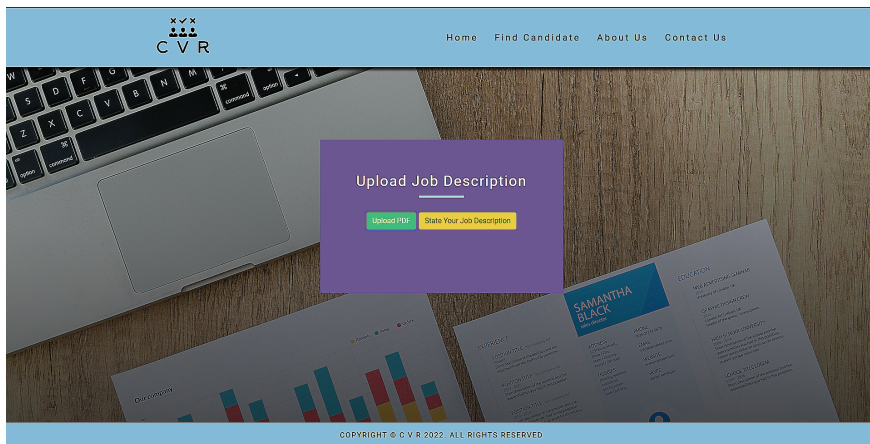


Fig. 7. Interface of uploading job description

the users should upload multiple CVs only in PDF format and a job description in PDF or text format.

As the final outcome, users will get the ranked recommended CVs as illustrated in Fig. 8. Moreover, the users have the flexibility to choose the number of CVs to be recommended and download them by clicking on their names.

Name	Email	Skills	Experience	Similarity Scores
AWSAF MAHMOOD	awsaf780@gmail.com	algorithms,android,arduino,c,c++,data analysis, data structures,django, docker,iot,java,javascript, machine learning,net working,node js,nosql,object-oriented programming,php,problem solving,python,qt,react js,rest api,sql,testbed	['North South University', 'Undergrad RA, Data Concentrator and System Integration', 'July 2021 – Present', 'Worked for a Government funded project to build a Cyber Physical Security Testbed', 'and implementation of Phasor Measurement Unit devices for the National Power Grid of', 'Bangladesh.', 'UNDERGRAD']	88.89%
ALKEMY HOSSAIN	alkemy48@gmail.com	algorithms,arduino,bootstrap,c,c++,css,data structures,figma,git,graphql,html,java,javascript,linux,machine learning,m	['Mentor', 'Deputy Event Manager', 'Mentored in the student mentoring class of the Department of Computer Science & Engi	66.67%

Fig. 8. Final recommendation

4 Result Analysis

4.1 Comparison and Selection of Similarity Metric

Six similarity checking metrics have been applied to compare which provides the more accurate and rational result. It was always found that conditional probability gives a better performance in comparison to other similarity metrics. Figure 9 shows the similarity scores of top 5 best-matched CVs for a job posting using the six metrics. The CVs shown were found using CP checker.

	Name	CP.Checker	Jaccard Score	Cosine Score	SC score	Dice score	Overlap score
62	MD ZAKARIA MASUD	83.33	11.36	24.73	75.07	20.41	83.33
12	ALKEMY HOSSAIN	66.67	11.11	32.59	63.25	20.00	66.67
33	GOLAM SALEHIN	66.67	26.67	21.53	60.17	42.11	66.67
38	KARNISH KHANAM	66.67	13.79	17.16	61.01	24.24	66.67
130	PRANAK BARUA	66.67	25.00	31.34	68.27	40.00	66.67

Fig. 9. Sorted similarity scores based on conditional probability

Among the applied metrics, in cosine similarity calculation, the frequency of a word in a text affects the score. However, to match skills between the CVs and

a job description, the frequency of the words is not essential. So, the scores it gives are not much high.

In soft cosine similarity calculation, the semantics of words are considered. However, semantics are ineffective in matching skills. As a result, the similarity scores are lower.

In case of Jaccard similarity, the union of the set of CV and job description text affects the score. Even if a resume includes all the skills mentioned in the job description, having additional skills will lower the similarity score.

Another metric, Dice similarity co-efficient or Sorensen-Dice index have similar effect as Jaccard similarity. The accuracy of dice similarity is slightly higher since the numerator contains twice the value of the overlapping segments of the two sets. However, the problem of eliminating candidates who have more than required skills still persists.

In case of Overlapping co-efficient, the score is affected by the minimum set of skills among a CV and a job description. This metric yields proper result when the job description has less skills compared to the CV; however, it does not provide the appropriate result when the job description has more skills. This again leads to the same problem as Dice similarity coefficient and Jaccard similarity.

The conditional probability algorithm, on the other hand, evaluates if the skills mentioned in a job description are present in a CV. As a result, additional skills in a resume that are not specified in the job description have no effect on the similarity score. As a result, it outperforms other similarity algorithms.

```
CV skills----- {'ajax', 'software testing', 'web design', 'json', 'django', 'communication', 'python', 'react js', 'android', 'html', 'github', 'jquery', 'rest api', 'javascript', 'bootstrap'}
JD Skills----- {'c', 'react js', 'asp.net', 'vue js', 'javascript', 'angular'}
Common Skills-- {'react js', 'javascript'}
```

Fig. 10. Matched skills using conditional probability

Due to the superior and rationality score, Conditional Probability score is chosen as the similarity metric for our system. Figure 11 depicts the similarity scores of the top five CVs using various similarity checking algorithms.

4.2 Selection of Recommendation Technique

Recommender systems are employed to find the best out of many choices. Initially, two techniques-Content-based recommendation and K Nearest Neighbor-were used to recommend top N CVs. Both the techniques take the best matched CV using Conditional probability. KNN algorithm produces recommendation results based on cosine similarity score. As cosine similarity does not yield higher similarity score due to vectorization and word frequency, the recommended CVs

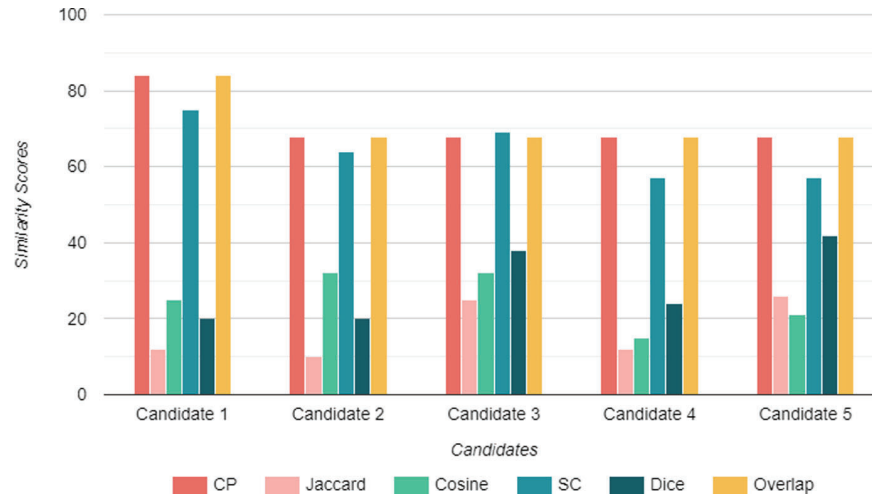


Fig. 11. Visualizing different similarity checking methods

are not suitable for the provided job description. On the other hand, the content-based recommendation technique recommends CVs using Conditional probability score which is more appropriate. Therefore, content-based recommendation has been implemented in the system for the recommendation purpose.

5 Conclusion

Hand-picking eligible candidates from a large number of CVs may be a challenging task in the recruitment process. Besides, certain important information may get unnoticed by the recruiter since there is no fixed format for writing CVs. To reduce the workload, a parser has been developed to extract important information from a CV, regardless of any particular format. Among the parsed information, skills of the candidates are matched with the skills extracted from a job description. Several similarity checking algorithms have been used to accomplish this out of which conditional probability checker provided with the best result. Finally, content-based recommendation is used to rank the top candidates for a job vacancy. The developed system will not only save the time of HR for shortlisting of eligible candidates but also get rid of the issue of working with various CV format. Currently, this work is restricted to computer science field only. However, future research has the potential to expand the work to other disciplines.

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